Vivekanand Education Society's Institute of Technology

An Autonomous Institute Affiliated to University of Mumbai

End Semester Examination
Special Exam 2023-24

Max marks: 60 | Duration: 2 hours
Branch: Computer Engineering | Semester: IV
Name of the Course: Operating System | QP Code: R23-CSC404_012023-24
Course code:CSC404

N.B. (1) Attempt any three out of the five questions.
(2) Figures to the right indicate full marks.
(3) Assume suitable data if necessary.

Marks

Q.1 (a) What are system calls of the operating system? Explain any 5 System calls? 10

(b) Describe the differences among short term, medium term and long term scheduling. 10

Q.2 (a) Consider the following set of processes, with length of CPU burst given in milliseconds as follows. 10

Process	Burst time	Arrival time	Priority
P1	8	0	3
P2	1	1	1
P3	3	2	2
P4	2	3	3
P5	6	4	4

(i)Draw the Gantt chart for FCFS,SJF and Preemptive priority scheduling?
(ii) What is the waiting and turn around time of each process for each of the above algorithm
(iii)Which algorithm results in minimum average waiting time?

(b) Explain the following in brief.(i) Process synchronization (ii) IPC 10

Q.3 (a) Explain synchronization problem in detail. How counting semaphore can be used to solve readers writers problem, 10

(b) Consider a system with 5 processes Po through P4.3 resource types A,B,C. 10
A= 10 instances
B= 5 instances
C= 7 instances
At time to, the following snapshot of the system has taken -

	Allocation			Maximum			Available		
	A	B	C	A	B	C	A	B	C
Po	0	1	0	7	5	3	3	3	2
P1	2	0	0	3	2	2			
P2	3	0	2	9	0	2			
P3	2	1	1	2	2	2			
P4	0	0	2	4	3	3			

Determine using Banker's algorithm whether the system is in safe state or not?

Q.4 (a) Explain the different allocation methods with reference to the file system. 10

(b) Explain the effect of page frame size on performance of page replacement algorithm. 10
Calculate page hit and miss for the following string using page replacement policies LRU and OPT. Page frame size is 4.
2,3,4,2,1,3,7,5,4,3,2,3,1

Q.5 (a) Suppose that a disk drive has 200 cylinders, numbered 0 to 199.The initial head position is at 100th track. The queue of pending requests in FIFO is 55,58,39,18,90,160,150,38,184.Calculate average seek time for each of the following algorithm. 10
(a) FCFS (b) SSTF (c) SCAN

(b) Write short notes on : (a) Real time operating system (b) Mobile operating system 10
© Network operating system (d) Embedded operating system

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